

JavaScript

DOM (Document Object Model):-

DOM - Document Object Model

- The **document object** represents the whole html document.

```
<form name="myform">
  <input type="text" name="firstname"/>
  <button type="button" onclick="validation()" > DOM </button>
</form>
```

```
<script type="text/javascript">
function validation()
{
  var fname=document.myform.firstname.value;
  alert("My name is: "+fname);
}
</script>
```

DOM Methods :-

Methods

document.getElementById()

document.getElementsByName()

document.getElementsByTagName()

1) document.getElementById –

document.getElementById()

- Return the element that has the ID attribute with the specified value.

```
<script type="text/javascript">
function show()
{
  var num=document.getElementById("number").value;
  alert(num);
}
</script>
```

```
<input type="text" id="number" />
<input type="button" value="submit" onclick="show()"/>
```

Example –

HTML Code –

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Document</title>

</head>

<body>

  <h1 id="h11">Welcome To JS1</h1>

  <h1>Welcome To JS2</h1>


  <h2 class="p11">I am Heading2</h2>

  <p class="p11">I am Paragraph</p>

  <h3 class="p11">I am Heading3</h3>


  <div>Box A</div>

  <div>Box B</div>

  <div>Box C</div>


  <script src="./dom.js"></script>


</body>

</html>
```

JS Code –

```
let a=document.getElementById("h11");

a.style.backgroundColor="red";
```

```
a.style.color="white";  
a.style.textAlign="center";
```

Output –

Welcome To JS1

Welcome To JS2

2) document.getElementsByName –



`document.getElementsByName()`

➤ Returns collection of all elements of particular document by name.

```
<script type="text/javascript">  
function show()  
{  
    var x=document.getElementsByName("abc");  
    alert("Total elements are:"+x.length);  
}  
</script>
```

```
<h4 name="abc">I</h4>  
<h4 name="abc">Am</h4>  
<h4 name="abc">Javascript</h4>  
  
<input type="button" value="submit" onclick="show()"/>
```

Example –

JS Code –

```
let b=document.getElementsByClassName("p11")  
for(let i=0;i<b.length;i++)  
{  
    b[0].style.color="green";  
    b[1].style.color="blue";  
    b[2].style.color="red";  
}
```

Output –

I am Heading2

I am Paragraph

I am Heading3

3) document.getElementsByTagName –



- Returns the collection of all the elements in the document with the given tag name.

```
<script type="text/javascript">
function show()
{
    var y=document.getElementsByTagName("h4");
    alert("Total elements are:"+y.length);
}
</script>
```

```
<h4>|</h4>
<h4>Am</h4>
<h4>Javascript</h4>
```

```
<input type="button" value="submit" onclick="show()"/>
```

Example –

JS Code –

```
for (let i=0;i<c.length;i++)
{
    c[0].style.fontSize ="50px";
    c[1].style.color ="purple";
    c[2].style.color ="yellow";
}
```

Output –

Box A

Box B

Box C

Events in JavaScript :-

- An event is an action that happens in the browser.
- Events can be triggered by user actions (click, key press, scroll) or by the browser itself (page load, resize).
- Events make web pages dynamic and interactive.
- Events are of different types like Mouse events, Keyboard events, Form events, Window events.

1) Keyboard Events –

- Keyboard events are triggered when the user interacts with the keyboard.
- These events help us detect key presses and perform actions based on user input.
- There are two main keyboard events in JavaScript: onkeydown & onkeyup.

keydown() & keyup()

```
<script>
$(document).ready(function(){
    $("input").keydown(function(){
        $("input").css("background-color", "green");
    });

    $("input").keyup(function(){
        $("input").css("background-color", "red");
    });
});
</script>
```

<input type="text">

Example –

HTML Code -

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Document</title>
```

```
</head>
```

```
<body>
```

```
  <input type="text" onkeydown="down()" onkeyup="up()">
```

```
<button onmouseover="over()" onmouseout="out()" onclick="(on)"
ondblclick="(ondbl)">Click</button>
```

```
<script src="./event.js"></script>
```

```
</body>
```

```
</html>
```

JS Code –

```
function down(){
```

```
    document.body.style.background="linear-gradient(white,purple)"
```

```
    document.body.style.height="100vh"
```

```
}
```

```
function up(){
```

```
    document.body.style.background="linear-gradient(purple,black)"
```

```
    document.body.style.height="100vh"
```

```
}
```

2) Mouse Events –

- Mouse events are triggered when the user interacts with the mouse.
- They help us detect clicks, movements, hovering, pressing, and releasing mouse buttons.
- Mouse events are very common for buttons, images, menus, and drag-drop actions



Some Events are:

- **onclick** - Fires when the user clicks once on an element. Used for buttons, links, etc.

click()

- When you click on an element, the click event occurs

```
<script>
$(document).ready(function(){
  $("button").click(function(){
    alert("I am click event");
  });
});
</script>
```

<button>Click here</button>

- **ondblclick** - Fires when the user double-clicks. Useful for opening/zooming actions.
- **onmouseover** - Fires when the mouse enters an element. Often used for hover effects.
- **onmouseout** - Fires when the mouse leaves an element. Used to remove hover effects.
- **onmousedown** - Fires when the mouse button is pressed down. Used for drag-start actions.
- **onmouseup** - Fires when the mouse button is released. Used to complete click or drag actions.

Example –

JS Code –

```
function over(){  
    document.body.style.background="radial-gradient(black,red,black)"  
    document.body.style.height="100vh"  
}
```

```
function out(){  
    document.body.style.background="radial-gradient(purple,blue,purple)"  
    document.body.style.height="100vh"  
}
```

```
function on(){  
    document.body.style.background="linear-gradient(white,purple,white,purple)";  
    document.body.style.height="100vh";  
}
```

```
function ondbl(){  
    document.body.style.background="linear-gradient(yellow,orange,red,brown,black)";  
    document.body.style.height="100vh";  
}
```

inner HTML :-



➤ This property is used to **set or return** the HTML content inside an element.

```
<script type="text/javascript">
function show()
{
    var data="<input type='text' name='name'><br><br><input
type='submit' value='Submit'>";
    document.getElementById('showdiv').innerHTML=data;
}
</script>
```

```
<input type="button" value="submit" onclick="show()"/>

<div id="showdiv"></div>
```

Example –

HTML Code -

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
    <title>Document</title>
```

```
</head>
```

```
<body>
```

```
    <script src="./method.js"></script>
```

```
</body>
```

```
</html>
```

JS Code –

```
let body=document.body
```

```
let div=document.createElement('div')
```

```
div.style.backgroundColor="silver"
```

```
div.style.color="purple"
```

```
div.style.borderStyle="dotted"
```

```
div.innerHTML +=
```

```
`<h1>Form</h1>
```

```
<form>
```

```
<input type="text" id="name" placeholder="Enter Your FirstName">
```

```
<br><br>
```

```
<input type="text" id="name" placeholder="Enter Your LastName">
```

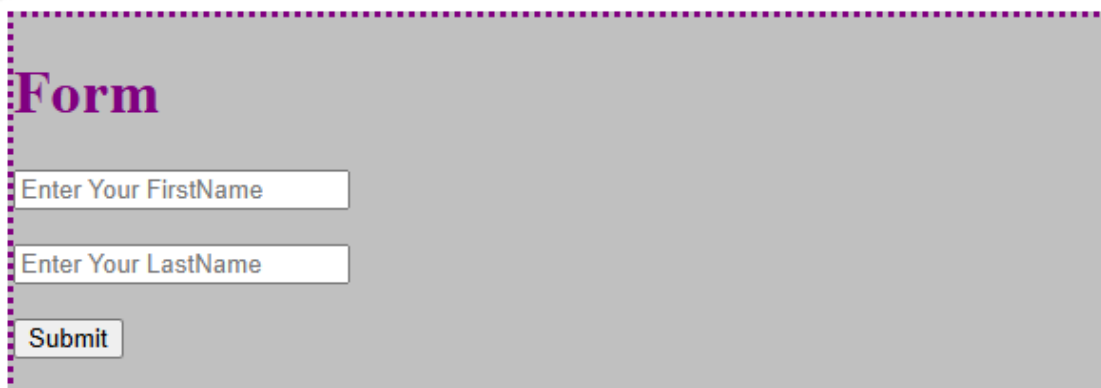
```
<br><br>
```

```
<button>Submit</button>
```

```
<br><br>`
```

```
body.appendChild(div)
```

Output –



Form